CLASS 8 — SCIENCE UNIT TEST SET A

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home Earth, a Unique Life-Sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. When an object is close to a concave mirror the image is: 1

(a) erect and diminished **(b)** inverted and diminished
(c) no image **(d)** erect and enlarged

2. The image formed by a convex mirror is always: 1

(a) inverted and diminished **(b)** erect and enlarged
(c) erect and diminished **(d)** inverted and enlarged

3. A woman walks towards a large concave mirror. She sees her inverted image get bigger and then: 1

(a) an inverted smaller image **(b)** a plane image
(c) no image **(d)** an erect, magnified image

4. The sidereal year is longer than the tropical year by about: 1

(a) 2 days **(b)** 20 hours **(c)** 20 minutes **(d)** 20 days

5. The extreme positions of sunrise occur on the: 1

(a) full moons **(b)** solstices **(c)** new moons **(d)** equinoxes

6. A lunar year falls short of the solar year by nearly: 1

(a) 30 days **(b)** 1 day **(c)** 100 days **(d)** 11 days

7. Organisms that eat only plants are: 1

(a) carnivores **(b)** omnivores **(c)** decomposers **(d)** herbivores

8. Growing the same crop repeatedly is called: 1

(a) crop rotation **(b)** mixed farming **(c)** monoculture **(d)** fallow farming

9. Decomposers are also called: 1

(a) autotrophs **(b)** predators **(c)** parasites **(d)** saprotrophs

10. The zygote is: 1

(a) the first cell formed after fertilisation
(b) a seed coat
(c) a sperm
(d) an egg

11. The greenhouse effect on Earth: 1

(a) removes the atmosphere **(b)** keeps Earth warm by trapping heat
(c) makes Earth colder **(d)** stops sunlight

12. In most mammals, the embryo develops: 1

(a) in water **(b)** in soil
(c) outside in eggs **(d)** inside the female's body

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. Differentiate concave and convex mirrors. 2

14. When does the Indian National Calendar year begin? 2

15. What are phases of the Moon? 2

16. What is a food chain? Give one example. 2

17. Define atmosphere, hydrosphere, geosphere and biosphere in one line each (any two). 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. What happens when a ray falls along the normal? Draw and explain. 3

19. Why are leap years skipped in 1700, 1800 and 1900 but not in 2000? 3

20. Differentiate producers, consumers and decomposers with examples. 3

21. Describe the development of a bird from fertilisation to hatching. 3

22. Explain the greenhouse effect and why it is important. 3

OR

Compare Venus and Earth in terms of greenhouse effect. 3

23. State three differences between reflection and refraction. 3

OR

A ray is incident on a plane mirror at 25° to the normal. Draw (describe) the reflected ray and find the angle of reflection and the angle between the reflected ray and the mirror. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. Describe artificial satellites: how they are launched, their types by use, and examples from India. 5

OR

Explain leap year rules in the Gregorian calendar with examples. 5

25. Explain sexual reproduction in flowering plants from pollination to seed formation. 5

OR

Discuss the triple planetary crisis and give actions to address it. 5

END OF QUESTION PAPER — **Total: 50 Marks**